

A Mechanistic Refusal Direction in a Small Instruct Model

The refusal decision is linearly legible in last-prompt-token activation *norms* (AUROC 0.98, leave-one-out 0.825, permutation $p=0.005$) where the first-token logit margin is not

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Abstract

Per the hexa-codex `cx_paper_gate` workflow rule, a paper ships only when every section claim reaches a 100% formal-blue or numerical-green verdict. We report the SAFETY-domain finding measured on a self-hosted full-precision `Qwen2.5-1.5B-Instruct` [5] (NVIDIA RTX 5070, HuggingFace `transformers` forward-pass hooks [4]): on a variance-controlled matched-pair adversarial-vs-benign set (20 adversarial + 20 benign-matched prompts, 20 refused / 20 answered by a fixed marker scan), the per-layer residual/attention/MLP activation-*norm* vector at the last prompt token (84 features = 28 layers $\times$ 3 sublayer kinds) *linearly separates* refused from answered. A difference-of-means refusal direction w yields a full-vector projection AUROC of 0.98; a leave-one-out (held-out) linear classifier reaches 0.825 (33/40) against a majority baseline of 0.50; and a 200-shuffle permutation test gives $p=0.005$ (0/200 label shuffles reach the observed accuracy). The same surface read through the first-token *logit* margin is a confirmed *negative* (the refused-vs-answered gap sits $5.9\times$ below the bimodality bar): the refusal decision is mechanistically legible in activation norms but *not* in the next-token logprob margin. The direction tracks the refusal *decision*, not the adversarial-vs-benign topic — the single adversarial-but-*answered* prompt projects to the answered side. **Scope (not claimed):** $n=40$ is small and the in-sample 84-dim AUROC can overfit; the load-bearing numbers are therefore the held-out leave-one-out accuracy and the permutation p , both reproduced by an independent recompute. We do *not* claim a single-direction ablation, a full sparse-autoencoder feature decomposition, or transfer to other models (those are deferred). The §Formula recompute (`verify/numerics_safety_refusal_direction.hexa`) passes 5/5 checks against the committed numbers-only matrix; verdict `.verdicts/sandbox/m4_safety_refusal_di`

1 Formula

SUPPORTED-NUMERICAL — the classifier is a *difference-of-means linear direction* in a standardized activation-norm feature space, the norm-space analogue of the linear refusal direction reported for activation *vectors* [1] and of representation reading more broadly [6]. Let $a_i \in \mathbb{R}^{84}$ be the last-prompt-token activation-norm vector for prompt i (feature j is the L_2 norm of one (layer, kind) hidden state, $\text{kind} \in \{\text{residual}, \text{attn}, \text{mlp}\}$), and let $y_i \in \{0, 1\}$ be its refusal label (1 = refused). Each feature is z -scored over the $n=40$ rows,

$$z_{ij} = \frac{a_{ij} - \mu_j}{\sigma_j}, \quad \mu_j = \frac{1}{n} \sum_i a_{ij}, \quad \sigma_j = \sqrt{\frac{1}{n} \sum_i (a_{ij} - \mu_j)^2}, \quad (1)$$

(population σ , with the degenerate $\sigma_j=0$ replaced by 1). The refusal direction and per-row projection score are

$$w_j = \text{mean}_{i: y_i=1} z_{ij} - \text{mean}_{i: y_i=0} z_{ij}, \quad s_i = \sum_{j=1}^{84} z_{ij} w_j, \quad (2)$$

and the refuse-vs-answer rule is $\hat{y}(a) = \mathbb{1}[w \cdot z(a) > \theta]$, with θ the midpoint of the two class-projection means. Three threshold-free / held-out statistics certify the direction [2, 3]: (A) the full-vector projection AUROC (rank statistic, refused = positive); (B) the leave-one-out held-out linear accuracy (refit w on the other $n-1$ rows, threshold at the train class-mean midpoint, predict the held-out row); (C) a 200-shuffle permutation p on the leave-one-out accuracy.

Recompute. The verifier `verify/numerics_safety_refusal_direction.hexa` reads the committed numbers-only matrix `.verdicts/sandbox/m2_safety_refusal_norms.tsv` (no model spawn, no GPU, no API) and recomputes w , the projections, and all three statistics, checking five conditions (all pass): (1) matrix shape 40×84 , balanced 20/20; (2) projection AUROC reproduces 0.98 (drift 0.0); (3) leave-one-out accuracy reproduces 0.825 (33/40, drift 0.0) and exceeds the majority 0.50; (4) permutation $p \leq 0.05$ (0/200 shuffles reach the observed accuracy, $p=0.00498$); (5) the topic-confound control — the lone adversarial-but-answered row projects to the answered side ($s = -25.81$, matching the original probe’s -25.81). Verdict `.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt` (5/5).

Permutation-test honesty (explicitly disclosed). The original probe shuffled labels with Python’s Mersenne-Twister (seed 1234); the hexa verifier uses its own fixed-seed linear-congruential Fisher–Yates shuffle for the same 200-permutation leave-one-out test. The test *design* is identical, the pseudo-random stream differs. The observed accuracy (0.825) sits so far above the chance band that 0/200 shuffles reach it under either generator, so $p=(0+1)/(200+1) \approx 0.005$. The p -value’s authority is the permutation design and the large margin, not a specific RNG implementation.

2 Method

SUPPORTED-NUMERICAL — activations are captured by the hook logic of `lm_foundry/tool/activation` (the HuggingFace `transformers` `register_forward_hook` backend), run on a self-hosted NVIDIA RTX 5070 (12 GB) host `ubu-1` serving *full-precision* (fp32) `Qwen2.5-1.5B-Instruct` [5] — the true weights, *not* a quantized GGUF — in a clean pinned virtual environment (`transformers==4.51.3`, `numpy<2`). For every prompt a forward hook on each of the 28 decoder layers records the L_2 norm of the residual-stream output, the self-attention output, and the MLP output *at the last prompt-token position* (the position whose hidden state forms the answer/refuse decision before generation), giving the ordered 84-feature vector. The refusal label is an exact reuse of the prior verdicts’ marker scan: a short 48-token greedy generation (`do_sample=False`, fp32) is labelled **refused** iff any of 24 fixed refusal markers (e.g. “i can’t”, “i’m sorry”, “i must decline”) appears in the first 200 characters — *no* LLM self-judge, per the SANDBOX honesty rule.

Test set (pre-registered, variance-controlled). The prompts are the matched-pair set of `bench/sandbox_stage4_refusal_bimodal_tighter.hexa` (HuggingFace private `dancinlab/hexa-codex-sandbox` adversarial sets default private per `cx_hf_safety_private`): 4 domains (social, physical-safety, self-harm, medical) $\times$ 5 pairs, each adversarial prompt paired with a benign prompt of the *same* domain, verb, approximate length, and prose-answer regime, so the only systematic adversarial–benign difference is the safety trigger. The observed label distribution is refused=20 / answered=20 (adv.refused=19/20, benign.refused=1/20) — clean specificity, with two honest label-flipped rows kept as the topic-confound control of §1. The committed analysis surface `.verdicts/sandbox/m2_safety_refusal_norms.tsv` contains the 40×84 activation norms and the binary labels *only*; the adversarial prompt strings are redacted per `cx_hf_safety_private`. A deterministic re-run of the forward pass reproduced the committed verdict’s label distribution and a crosscheck projection AUROC of 0.98 bit-for-bit before the matrix was dumped, so the

committed numbers are faithful to the original `motif_summary.json`. All capture ran at \$0 incremental on the self-hosted GPU. Verdict header: `.verdicts/sandbox/m2_safety_mechanical_probe.txt`.

3 Benchmark

SUPPORTED-NUMERICAL — the real bench is the 40-row $\times$ 84-feature separability analysis recomputed end-to-end from the committed matrix. The headline statistics (verbatim from the recompute verdict) are

statistic	value	baseline	verdict link
full-vector projection AUROC	0.98	0.50 (chance)	recompute (5/5)
leave-one-out held-out accuracy	0.825 (33/40)	0.50 (majority)	recompute (5/5)
permutation p (200 shuffles)	0.005 (0/200)	—	recompute (5/5)
adv-but-answered row, projection	−25.81	thr $\approx$ 0	answered-side OK

The separability is carried by three mechanistically distinct activation-norm motifs (single-feature AUROC and Cohen’s d , verbatim from the probe verdict):

motif	layer band	top AUROC	Cohen’s d	sign
mid-layer residual amplification	L17–L19	0.953 (L19)	+2.56	refused > answered
mid-layer MLP amplification	L17–L22	0.930 (L18)	+2.22	refused > answered
late-layer attention suppression	L22/L23/L26	0.968 (L23, inv.)	−2.54	refused < answered

These differ in sublayer (residual readout vs MLP write vs attention output), sign (amplification vs suppression), and layer band (mid vs late), so the linear direction is not a single-feature artifact. The in-sample 84-dim projection AUROC (0.98) can overfit at 84 features / 40 samples — which is exactly why the load-bearing numbers are the held-out leave-one-out accuracy (0.825) and the permutation p (0.005), both reproduced by the independent recompute. Verdict: `.verdicts/sandbox/m2_safety_mechanical_probe.txt` + the recompute `.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt`.

4 Benefit

SUPPORTED-NUMERICAL — the finding lands three paper-grade, interpretability-actionable deltas:

- **The activation-norm surface succeeds where the first-token logit margin fails — a measured route-vs-route delta.** On the *same* matched-pair set, the route(a) probe (the next-token top1–top2 logprob margin) is a confirmed *negative*: refused and answered cluster means converge to 1.73 vs 1.32 logprob (gap 0.40), which is $5.9\times$ *below* the bimodality bar (gap 0.40 vs the $2\sigma_{\max}$ threshold of 2.40), so `safety_signal_present=false` (`.verdicts/sandbox/m2_safety_bimodality_tighter.txt`). The activation-norm direction, on the identical labels, reaches held-out 0.825 (permutation $p=0.005$). **The refusal decision is legible mechanistically but not in the next-token margin** — so an activation tap, not a logit/logprob probe, is the right instrument; this directly motivated routing the SANDBOX safety lane to the `transformers` forward-hook backend rather than the `llama-server` logprob API.

- **The direction tracks the refusal *decision*, not the adversarial-vs-benign topic.** The lone adversarial-topic-but-*answered* prompt (an acetaminophen-dose question the model chose to answer) projects to the *answered* side ($s = -25.81$, threshold ≈ 0), re-computed identically by the verifier. A direction that merely encoded “adversarial topic” would have placed it on the refused side; this control is the difference between a refusal probe and a topic classifier.
- **An honest, recompute-verifiable interpretability surface at \$0.** The entire finding is reproducible from a 46 KB committed numbers-only matrix and one hexa verifier — no GPU, no model download, no adversarial prompt text leaves the private set. This converts a previously $\bigcirc$ speculation-fenced empirical probe into a $\bullet$ numerical-green recompute surface, which is precisely the `cx_paper_gate` condition this paper had to meet.

All three benefits are measured at \$0 incremental cost on a self-hosted GPU; the recompute surface — which closes the prior cycle’s `BLOCKED_PENDING_FORMULA_RECOMPUTE` residual — is part of the contribution.

5 Verdict matrix

Per `cx_paper_sections`, every section claim links to its verdict file. All four rows read $\bullet$ `SUPPORTED-NUMERICAL`.

Table 1: SAFETY paper verdict matrix (4 sections $\times$ g5 rubric tier).

section	verdict file
§1 formula	<code>.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt</code> (linear-direction recom)
§2 method	<code>.verdicts/sandbox/m2_safety_mechanical_probe.txt</code> (capture protocol header) + recom
§3 benchmark	<code>.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt</code> (AUROC/LOO/ $p + 3$)
§4 benefit	<code>.verdicts/sandbox/m2_safety_bimodality_tighter.txt</code> (route(a) negative contrast) + r

Ship condition. All four rows graduate to $\bullet$ `SUPPORTED-NUMERICAL`. `cx_paper_significance` holds: a closed-form classifier (difference-of-means linear refusal direction) + a real 40×84 activation-norm benchmark + quantified benefit (held-out 0.825 vs majority 0.50; the $5.9\times$ -below-bar route(a) negative; the topic-confound control). No row remains $\bigcirc$ `INSUFFICIENT` per `cx_paper_violation`.

6 Limitations and honest caveats

- **$n=40$ is small; the in-sample AUROC can overfit.** With 84 features and 40 samples the full-vector projection AUROC (0.98) is in-sample and over-optimistic. Only the held-out leave-one-out accuracy (0.825) and the permutation p (0.005) are treated as load-bearing; both are reproduced by the independent recompute.
- **Activation *norms*, not full vectors or SAE features.** The committed surface is the per-(layer, kind) L_2 norm, not the dense hidden state. This establishes that a refusal *direction* exists in norm space; it is *not* a sparse-autoencoder feature atlas nor a causal single-direction ablation [1]. Those are deferred to M5.SAFETY.
- **Marker-scan label has a known 1-row noise floor.** One benign medical prompt (ER warning-signs) tripped a refusal marker; the AUROC, LOO, and p are reported against that exact label set for honesty, not a cleaned one.

- **Single model, single feature site.** Qwen2.5-1.5B-Instruct fp32, last prompt-token only. Transfer to other models, sizes, or token positions is unmeasured.
- **Publish-lint (g51) residual.** This 4-section draft does not yet meet the commons ≥ 10 -page + fal.ai-figure publish bar; that is a separate **M5.SAFETY-release** condition, deliberately not padded here.

7 Discussion and outlook

The refusal decision in a small instruct model is linearly readable from mid-layer residual/MLP amplification and late-layer attention suppression at the last prompt token, even though it is invisible in the first-token logit margin. The single most important follow-up is a *causal* test: ablate the recomputed direction w (or its dense-vector analogue) at inference and measure the change in refusal rate on held-out adversarial prompts — upgrading the present correlational direction to an interventional one. A full sparse-autoencoder decomposition of the three motifs (the M5.SAFETY release target) would replace the norm-space proxy with named features.

Reproducibility

Code: <https://github.com/dancinlab/hexa-codex>. Capture hook: `lm_foundry/tool/activation_capture`.
 $\S$ Formula recompute: `verify/numerics_safety_refusal_direction.hexa` (\$0 local, no GPU;
`HEXA_SHIM_NO_DARWIN_LANDING=1 RESOURCE_LOCAL_HEXA=1 HEXA_CODEX_ROOT="$PWD" hexa run`
`verify/numerics_safety_refusal_direction.hexa`). Committed numbers-only matrix: `.verdicts/sandbox`
(adversarial prompt text redacted per `cx_hf_safety_private`). Verdicts under `.verdicts/sandbox/`.
Build: `make (pdflatex  $\times$  3 + bibtex)`.

References

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- [3] Phillip I. Good. *Permutation Tests: A Practical Guide to Resampling Methods for Testing Hypotheses*. Springer, 2000. doi: 10.1007/978-1-4757-3235-1.
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